Conic Sections

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

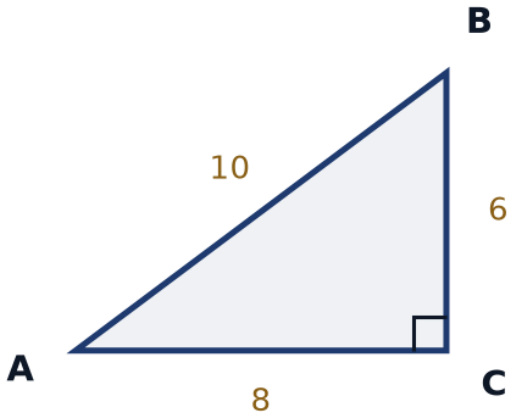
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(2)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for trigonometric functions in the coordinate plane consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

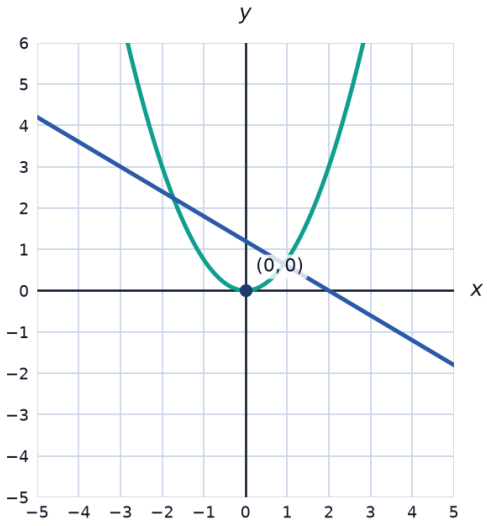
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Conic Sections & Analytic Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Identify the conic section $4x^2 - 9y^2 - 16x - 18y - 29 = 0$ by completing the square. Find its center, vertices, foci, and asymptotes.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Eliminate the parameter t from the parametric curve $x(t) = 3 + 4\cos t$, $y(t) = -2 + 5\sin t$ for $0 \le t \le 2\pi$. Identify the conic section, its center, and its vertices.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Evaluate the exact value of $\cos\left(\arcsin\left(-\frac{3}{5}\right) + \arctan(2)\right)$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · AdvancedFind the matrix R that rotates vectors by $\theta = \frac{2\pi}{3}$ radians counterclockwise, and use it to find the image of the vector $\vec{w} = (4;\ -2)$. Verify that the magnitude of $\vec{w}$ is preserved under the transformation.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For what value of k will the system have no solution?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

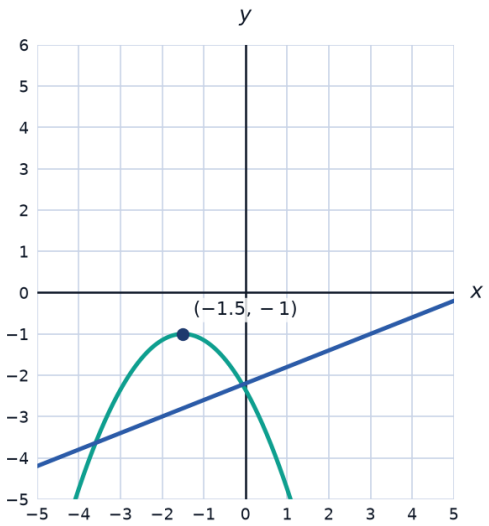
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the exact value of $\cos\left(\frac{7\pi}{12}\right)$ and $\sin\left(\frac{7\pi}{12}\right)$ using angle addition identities.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Polynomial Functions

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Rational Expressions

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find the area enclosed between $y = x^2$ and $y = x + 2$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Sequences & Series

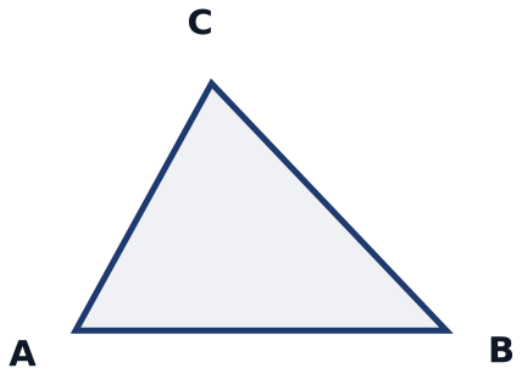
Find all solutions to $2\cos^2\theta - \cos\theta - 1 = 0$ on the interval $[0, 2\pi)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 102°

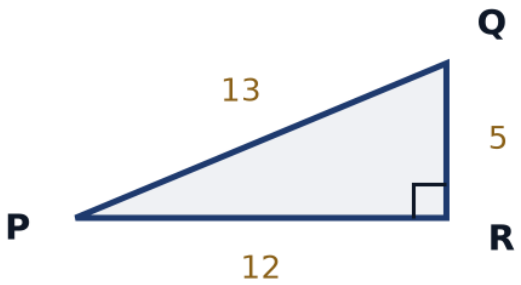
b. 112°

c. 94°

d. 147°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying special right triangles and exact value trigonometric ratios, $m\angle A = 34^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 101°

b. 111°

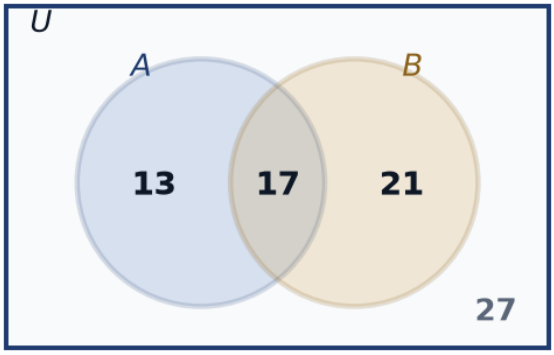
c. 93°

d. 146°

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for trigonometric functions on the unit circle consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?



a. 17.5

b. 18.5

c. 19.5

d. 21.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 4$ in trigonometric functions on the unit circle, evaluate $f(4)$.

a. 20

b. 24

c. 16

d. 24

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for evaluating trigonometric functions: exact values consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 5$ in evaluating trigonometric functions: exact values, evaluate $f(4)$.

a. 25

b. 30

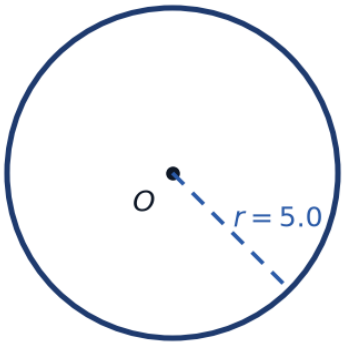
c. 20

d. 29

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation of the circle with center $(3, -1)$ and radius 5.



a. $(x+3)^2 + (y-1)^2 = 25$

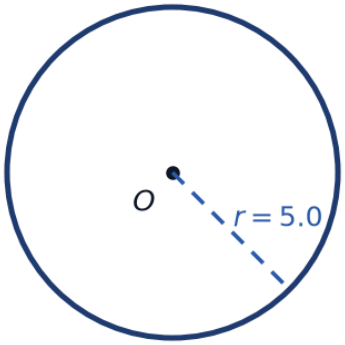
b. $(x-3)^2 + (y+1)^2 = 25$

c. $(x-3)^2 + (y-1)^2 = 5$

d. $x^2 + y^2 = 25$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find where the circle $(x-3)^2 + (y-4)^2 = 25$ crosses the x-axis.



a. $(3, 0)$ only

b. $(-3, 0)$ and $(6, 0)$

c. No intersection

d. $(0, 0)$ and $(6, 0)$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A circle $\mathcal{C}$ with center $C(0, k)$ on the positive y-axis passes through the focus F of the parabola $x^2 = 4y$. If $\mathcal{C}$ is tangent to the parabola at two distinct points, P and Q, what is the area of quadrilateral $PFQC$?

a. $8\sqrt{3}$

b. $4\sqrt{3}$

c. $16\sqrt{3}$

d. $6\sqrt{3}$

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The equation of an ellipse is given by: $9x^2 + 4y^2 - 36x + 8y + k = 0$ If the length of the major axis of the ellipse is 6, what is the value of the constant k?

a. 4

b. -31

c. -41

d. -104

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

An ellipse $\mathcal{E}$ and a hyperbola $\mathcal{H}$ are confocal, sharing the foci $F_1(-4, 0)$ and $F_2(4, 0)$. The eccentricity of $\mathcal{E}$ is $\frac{4}{5}$, and the eccentricity of $\mathcal{H}$ is 2. If P is an intersection point of $\mathcal{E}$ and $\mathcal{H}$ in the first quadrant, what is the area of ΔPF_1F_2?

a. $6\sqrt{3}$

b. $12\sqrt{3}$

c. $3\sqrt{3}$

d. $4\sqrt{5}$

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A parabola has its vertex at the origin $(0, 0)$ and passes through the point $(2, -4)$. If the directrix of the parabola is a horizontal line, what is the equation of the directrix?

a. y = 4

b. y = -4

c. x = 4

d. x = -4

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circle has its center at the point $(2, -3)$ and a radius of 5 units. Which of the following is the standard form equation of this circle?

a. $(x-2)^2+(y+3)^2=25$

b. $(x+2)^2+(y-3)^2=5$

c. $(x-2)^2+(y+3)^2=5$

d. $(x+2)^2+(y-3)^2=25$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Derive the reflection property: light from one focus of an ellipse reflects to the other.

a. By Snell's law variant: angle of incidence = angle of reflection using the definition $PF_1 + PF_2 = 2a$ and tangent bisects angle F_1PF_2

b. Only for circles

c. Only for parabolas

d. Arbitrary property

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Prove that the chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ on $y^2 = 4ax$ has equation $y(t_1 + t_2) = 2x + 2at_1t_2$.

a. Use calculus

b. Incorrect formula

c. Slope $= 2/(t_1 + t_2)$; using point-slope through $(at_1^2, 2at_1)$ gives the result

d. Only for $t_1 = t_2$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the equation of the hyperbola with asymptotes $y = \pm 2x$ and passing through $(3, 4)$.

a. $\frac{x^2}{5} - \frac{y^2}{20} = 1$

b. $y^2 - 4x^2 = 0$

c. $4x^2 - y^2 = 32$

d. $x^2 - 4y^2 = 4$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A chord PQ of the parabola $y^2 = 4ax$ passes through the focus. Prove that $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$ where S is the focus.

a. Not always true

b. Using focal chord parametrization $t_1 t_2 = -1$ and $SP = a(1 + t_1^2)$

c. Only for the latus rectum

d. Use AM-GM

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Derive the eccentricity of a conic from the focus-directrix ratio definition $PF/PD = e$ for the parabola.

a. $e = 2$ for parabola

b. For parabola, $e = 1$: distance to focus equals distance to directrix

c. $e > 1$ for parabola

d. $e < 1$ always

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find the values of k so that the line $y = kx - 1$ is tangent to the parabola $x^2 = 4y$.

a. $k = 0$

b. $k = \pm 1$

c. Any k

d. $k = 2$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Sequences & Series

Prove that the difference of the focal radii of a point on a hyperbola equals $\pm 2a$.

a. From definition $|PF_1 - PF_2| = 2a$, which defines the hyperbola

b. Only for $e > 2$

c. $PF_1 - PF_2 = 2c$

d. It equals $2b$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Trigonometric Functions

For the ellipse $b^2x^2 + a^2y^2 = a^2b^2$, confirm this equals standard form.

a. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

b. $x^2 + y^2 = a^2 + b^2$

c. $ax + by = ab$

d. $b^2 + a^2 = 1$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Functions & Their Graphs

Find the polar equation of an ellipse with focus at the origin, semi-major $a = 5$, $e = 0.6$.

a. $r = \frac{5}{1 - \cos\theta}$

b. $r = \frac{5}{1 + 0.6\cos\theta}$

c. $r = \frac{3.2}{1 - 0.6\cos\theta}$

d. $r = \frac{8}{1 + 0.6\cos\theta}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Matrices & Systems

Convert the hyperbola $4x^2 - 9y^2 = 36$ to standard form.

a. $\frac{x^2}{9} - \frac{y^2}{4} = 1$

b. $x^2 - y^2 = 9$

c. $\frac{x^2}{36} - \frac{y^2}{36} = 1$

d. $\frac{x^2}{4} - \frac{y^2}{9} = 1$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability & Statistics

Find the equation of the ellipse with foci $(\pm 3, 0)$ and sum of focal radii $= 10$.

a. $\frac{x^2}{10} + \frac{y^2}{9} = 1$

b. $\frac{x^2}{9} + \frac{y^2}{16} = 1$

c. $\frac{x^2}{25} + \frac{y^2}{16} = 1$

d. $x^2/16 + y^2/9 = 1$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the equation of the circle with diameter endpoints $(2, 3)$ and $(8, 7)$.

a. $(x-5)^2 + (y-5)^2 = 13$

b. $(x-5)^2 + (y-5)^2 = 26$

c. $(x-5)^2 + (y-5)^2 = 25$

d. $x^2 + y^2 = 13$

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